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Arshia Razavi

Summary

MSc Physics graduate (2026) applying stochastic modeling, simulation, and machine learning to large datasets in Python and SQL; builds and evaluates data and AI systems at AltaML. Studying for CFA Level I.

Education

University of Calgary, M.Sc. Physics — Alberta, Canada
Thesis: *Learning and Stabilizing Memories in Recurrent Spiking Neural Networks.* Simulated spiking networks and analyzed large-scale Allen Institute neural recordings (AllenSDK, Python) — Completed Jul 2026

Sharif University of Technology, B.Sc. Physics — Tehran, Iran
Thesis: *Hybrid-Type Synchronization Transitions in Complex Networks* — 2017–2022

Experience

AltaML, Full Stack Software Engineer — Alberta, Canada
Python, FastAPI, PostgreSQL, LLMs, GCP, Azure, Next.js — Apr 2026 – Present

- Led an LLM classifier's model migration with an evaluation harness; fixed a prompt weakness that dropped the new model to 98.0%, restoring **99.7% exact match on 397 files**.
- Built a borrower portal for a **lending platform** that collects and validates financial documents: Next.js frontend, FastAPI + PostgreSQL backend, secure authentication, configurable validation rules.
- Determined which campaign objective/optimization pairs Meta's API accepts by testing them against the live API, unblocking pairs our validation wrongly rejected; added spend-share metrics to optimization recommendations.

AltaML, Associate Machine Learning Developer (Intern) — Alberta, Canada
Python, FastAPI, SQL, pandas, GPT-4o, Arize, Azure • 42 merged PRs — Sep 2025 – Apr 2026

- Raised LLM extraction accuracy on **financial spreadsheets from 46% to 88%** field-level correctness by feeding cached formula values (openpyxl) into GPT-4o prompts, measuring each change with Arize evaluation runs.
- Built a deterministic numeric-comparison engine for **financial variance checks** (tolerance-based equality, threshold flags; 49 unit tests) feeding an automated DOCX reporting pipeline.
- Developed FastAPI/SQL backend services for a document-review platform, replacing one query per assessment with a single bulk aggregate query and adding batch processing that rolls back on failure.

Turing, Physics Data Analyst — Remote, California, US
Designed physics problem datasets targeting known reasoning failure modes in Gemini — Aug 2025 – Dec 2025

Skills

- **Quantitative:** Probability, stochastic processes, statistical inference, time-series analysis, Monte Carlo
- **Programming:** Python (NumPy, pandas, SciPy, scikit-learn), SQL, Excel (formulas, charts), C++, MATLAB
- **ML & Data:** Regression, classification, optimization, PyTorch, LLM evaluation (Arize, MLflow)
- **Cloud & Tools:** Azure, GCP, AWS, Docker, FastAPI, Git, GitHub Actions, LaTeX

Honors & Awards

Silver Medal, International Physics Olympiad (IPhO 2017); Gold, Iranian National Physics Olympiad — 2016–17
Winner Team, QAIventure GenQ Hackathon, quantum reservoir computing for load forecasting — 2025
Scholarships: Alberta Graduate Excellence (11,000 CAD); UCalgary and Sharif tuition awards — 2017–2025